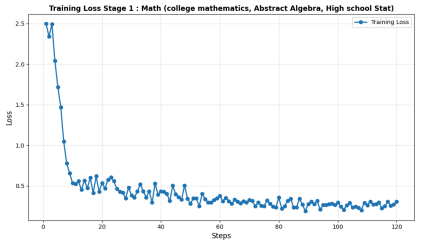
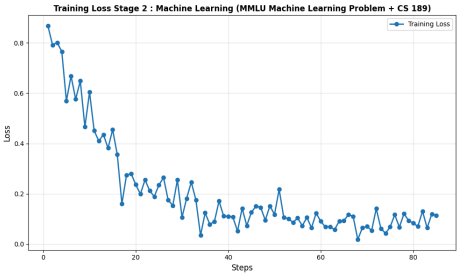
Homework 5 Coding: Finetuning LLMs with SFT

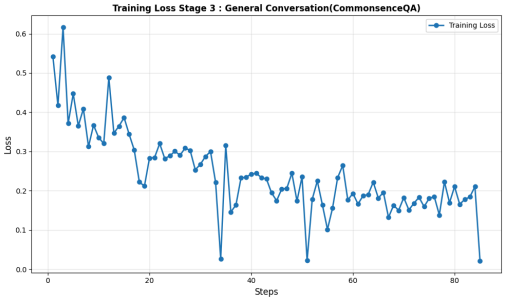
1. What techniques did you use to train your model? Include a detailed explanation of your final methods, including your loss curves, hyperparameters, datasets, optimizers, etc. Make sure to explain what each dataset contains, not just its name and cite where you got it from. If you took techniques from existing works, please cite those as well. (3-6 sentences)
    a. I fine-tuned Qwen/Qwen2.5-0.5B-Instruct with TRL's SFTTrainer using token-level cross-entropy and the 8-bit AdamW optimizer (adamw_8bit) for full-parameter SFT.
    b. Training used a 3-stage curriculum: (1) math from Hugging Face TIGER-Lab/MMLU-STEM subjects (college_mathematics, abstract_algebra, high_school_statistics), (2) machine learning from MMLU-STEM (machine_learning) plus a CS189-style MCQ CSV (questions , A–E options, answer letter) concatenated repeatedly, and (3) general capability from tau/commonsense_qa (CommonsenseQA) converted into 5-choice format and mixed with a replay subset of the ML data to reduce forgetting.
    c. My main hyperparameters were batch size 4, gradient accumulation 4, LR = 1e-5, 5 epochs per stage, warmup ratio 0.03, weight decay 0.01, and a cosine LR scheduler. For monitoring, I plotted training/validation loss curves by extracting each trainer's log_history and graphing loss vs. step separately for Stage 1/2/3 (Math, ML+CS189, CommonsenseQA). For the prediction, I used TTA (Test-Time Augmentation) with a majority vote.
    d. This is a loss graph for first stage (TIGER-Lab/MMLU-STEM subjects)



    e. Second stage (machine learning from MMLU-STEM CS189-style MCQ CSV)



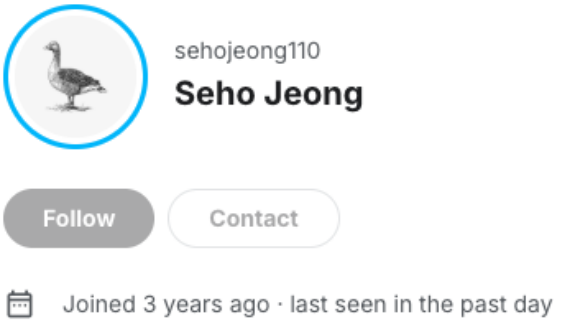
    f. This is a loss graph for third stage (CommonsenseQA)



2. What did you try that didn't work, and why do you think it failed? (1-2 sentences)
    a. I initially used a multi-stage curriculum to cover several abilities (college math, abstract algebra, statistics, and machine learning) in separate fine-tuning stages. However, because each dataset was relatively small and the model was only 0.5B, even with replay the later stages caused strong catastrophic forgetting and the model lost performance on earlier domains.
3. Provide 2 different examples of catastrophic forgetting (something the model could do before but can't do now). Include the input, the outputs from the base model and your finetuned model, and one sentence on why you think the model has degraded for this input. Note that you don't need to show examples from the provided datasets; you can use any type of input. (2-3 sentences)
    a. Example 1 (mmlu_pro_00001, programming trace): Input: "In the program below, y is a positive integer …" Base model output: \boxed{A} (correct) → Finetuned output: \boxed{E} (incorrect). I think this is catastrophic forgetting because later-stage fine tuning shifts the model toward a different answer distribution/heuristic, so it loses the earlier "step-by-step trace" behavior it had for this kind of code question.
    b. Example 2 (mmlu_pro_00010, ML/NLP concept): Input: "Statement 1 | The Stanford Sentiment Treebank …" Base model output: \boxed{A} (correct) → Finetuned output: \boxed{B} (incorrect). I think it degraded because the final fine tuning overwrote parameters useful for this specific concept question (classic interference when training sequentially on narrow datasets).
4. Provide 2 different examples of non-CS189 problems that your finetuned model gets right and your base model gets wrong (i.e. what unintended skills did your model gain). Include the input, the outputs from the base model and your finetuned model, and one sentence on why you suspect the model has gained this ability. (2-3 sentences)
    a. Case 1 (mmlu_machine_learning_00009) — Input: "As the number of training examples goes to infinity …" (True answer A). The base model predicted \boxed{C} (wrong), but the fine-tuned model predicted \boxed{A} (correct).
    Why it improved: SFT on ML-style questions likely strengthened the model's "generalization/asymptotics" intuition (train vs. validation behavior as data grows).
    b. Case 2 (mmlu_machine_learning_00005) — Input: "Statement 1 | The log-likelihood of the data …" (True answer B). The base model predicted \boxed{A} (wrong), but the fine-tuned model predicted \boxed{B} (correct).
    Why it improved: Repeated exposure to probability/likelihood-style reasoning during fine-tuning can "calibrate" these standard ML/statistics concepts better than the base instruction model.

Kaggle Submission

1. Username



2. Kaggle Leaderboard

